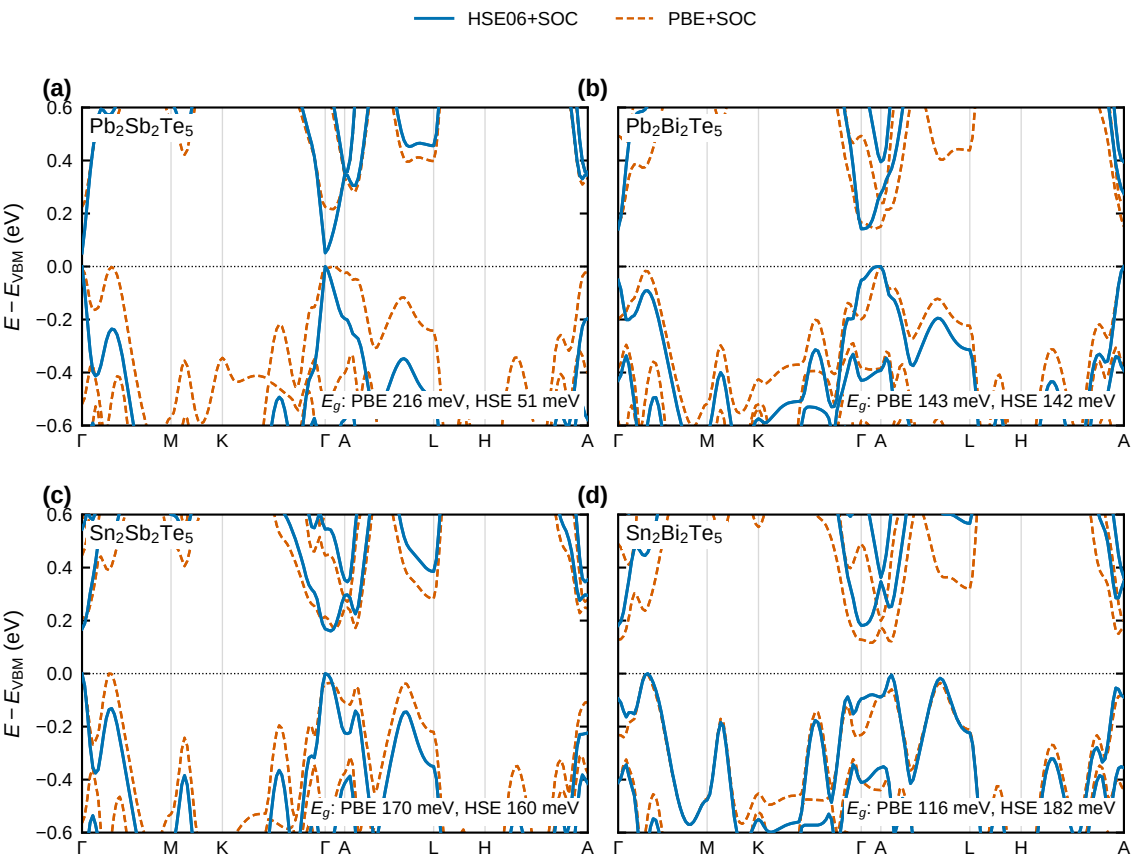

HSE06+SOC
PBE+SOC
(a)
Pb2Sb2Te5
Eg: PBE 216 meV, HSE 51 meV
(b)
Pb2Bi2Te5
Eg: PBE 143 meV, HSE 142 meV
(c)
Sn2Sb2Te5
Eg: PBE 170 meV, HSE 160 meV
(d)
Sn2Bi2Te5
Eg: PBE 116 meV, HSE 182 meV
E − EVBM (eV)
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
Γ
M
K
Γ
A
L
H
A